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Stern drives having breakaway lower gearcase

Abstract

A stern drive is for propelling a marine vessel in water. The stern drive has an upper drive unit with a lower mounting surface; a lower gearcase coupled to the lower mounting surface and a trailing end surface that is angled relative to the lower mounting surface; and a propeller shaft extending forwardly from the lower gearcase and being configured to rotate a propeller for pulling the marine vessel in the water. The upper drive unit and the lower gearcase are configured such that when a forward side of the lower gearcase impacts an underwater obstruction, the lower gearcase is caused to pivot relative to the upper drive unit until the trailing end surface impacts the lower mounting surface, which thereby causes the lower gearcase to completely uncouple from the upper drive unit.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) The present application is a continuation of U.S. application Ser. No. 17/518,935, filed Nov. 4, 2021, which is a continuation of U.S. application Ser. No. 16/909,082, filed Jun. 23, 2020, now U.S. Pat. No. 11,208,190, issued Dec. 28, 2021, which applications are hereby incorporated by reference in their entirety.

FIELD

(1) The present disclosure relates to stern drives for marine vessels.

BACKGROUND

(2) The following U.S. Patents are incorporated herein by reference in entirety:

(3) U.S. Pat. No. 7,234,983 discloses a marine vessel and drive combination having port and starboard tunnels formed in a marine vessel hull raising port and starboard steerable marine propulsion devices to protective positions relative to the keel.

(4) U.S. Pat. No. 7,435,147 discloses a marine propulsion device provided with a breakaway skeg having first and second attachment points. The first and second attachment points are configured to result in the second attachment points disengaging from a gear case or housing structure prior to the first attachment point. The attachment points can comprise open or closed slots and, when an open slot is used for the first attachment point, it can be provided with a first edge along which a first pin can exert a force along a preselected angle in response to an impact force on the skeg. The arrangement of attachment points allows a reaction force at the second pin to be predetermined in a way that assures the detachment of the skeg from the housing structure prior to the detachment of the housing structure from another structure, such as the boat hull, or transom.

(5) U.S. Pat. No. 7,867,046 discloses a marine drive having a break-away mount provided by hollowed-out threaded fasteners mounting first and second sections of the drive and breaking away in response to a given underwater impact against the second section to protect the first section and the vessel.

(6) U.S. Pat. No. 8,011,983 discloses a marine drive having a break-away mount mounting first and second sections of the drive and breaking-away in response to a given underwater impact against the second section to protect the first section and the vessel.

(7) U.S. Pat. No. 9,481,439 discloses a stern drive for a marine vessel. The stern drive comprises a gimbal housing that is configured for connection to the marine vessel, a gimbal ring that is steerable with respect to the gimbal housing about a vertical steering axis, a driveshaft housing that is connected to the gimbal ring, and a trim actuator that is configured to trim the driveshaft housing about a horizontal trim axis. The trim actuator has a first end that is pivotably connected to the gimbal ring at a horizontal first pivot axis and a second end that is pivotably connected to the driveshaft housing at a horizontal second pivot axis. A resilient driveshaft housing vibration isolator is located along the second pivot axis. The resilient vibration isolator isolates vibration forces on the driveshaft housing. A resilient gimbal ring vibration isolator is located along the trim axis. The gimbal ring vibration isolator isolates vibration forces on the gimbal ring.

SUMMARY

(8) This Summary is provided to introduce a selection of concepts that are further described below in the Detailed Description. This Summary is not intended to identify key or essential features of the claimed subject matter, nor is it intended to be used as an aid in limiting the scope of the claimed subject matter.

(9) In certain examples disclosed herein, a stern drive is for propelling a marine vessel in water. The stern drive has an upper drive unit with a lower mounting surface; a lower gearcase coupled to

the lower mounting surface and having a trailing end surface that is angled relative to the lower mounting surface; and a propeller shaft extending forwardly from the lower gearcase and being configured to rotate at least one propeller for pulling the marine vessel in the water. The upper drive unit and the lower gearcase are configured such that when a forward side of the lower gearcase impacts an underwater obstruction, the lower gearcase is caused to rearwardly pivot relative to the upper drive unit until the trailing end surface impacts the lower mounting surface, which thereby causes the lower gearcase to completely uncouple from the upper drive unit.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The present disclosure is described with reference to the following Figures. The same numbers are used throughout the Figures to reference like features and like components.
- (2) FIG. 1 is a perspective view of a stern drive according to the present disclosure.
- (3) FIG. 2 is an exploded view looking down at the upper drive unit and lower gearcase of the stern drive.
- (4) FIG. 3 is an exploded view looking up at the upper drive unit and lower gearcase.
- (5) FIG. 4 is a side view of the stern drive as it impacts an underwater obstruction.
- (6) FIG. 5 is a side view of the stern drive immediately after it impacts an underwater obstruction, showing how the lower gearcase pivots rearwardly relative to the upper drive unit until a trailing end surface of the lower gearcase impacts a lower mounting surface of the upper driver unit.
- (7) FIG. 6 is a side view of the lower gearcase separated from the upper drive unit after impact of the trailing end surface on the lower mounting surface and uncoupling of the lower gearcase from the upper drive unit.

DETAILED DESCRIPTION

- (8) FIGS. 1-3 depict a stern drive **10** according to the present disclosure. The stern drive **10** extends from top **12** to bottom **14** in an axial direction **16**, from forward side **18** to rearward or trailing side **20** in a longitudinal direction **22** that is transverse to the axial direction **16**, and from port side (see FIG. 1) to starboard side (the opposite side of FIG. 1, not shown) in a lateral direction **28** that is transverse to the axial direction **16** transverse to the longitudinal direction **22**.
- (9) The stern drive **10** has an upper drive unit **30**, which in use is coupled to the transom of a marine vessel **31** (see FIG. 4). The stern drive **10** also has a lower gearcase **32** that depends from the upper drive unit **30**. The upper drive unit **30** generally includes a driveshaft housing **34** and an adapter plate **36**. In examples not shown, the driveshaft housing **34** and adapter plate **36** could instead be more than two components or could instead be formed together as a monolithic component. In the illustrated example, as shown in dash-and-dot lines in FIGS. 2 and 3, the adapter plate **36** is sandwiched between the upper drive unit **30** and the lower gearcase **32**. The driveshaft housing **34** has a lower mounting flange **38** (see FIG. 3) on which the adapter plate **36** is mounted. The adapter plate **36** has an upper mounting surface **40** (see FIG. 2) that faces and is coupled to the lower mounting flange **38**, as will be further described herein below. The adapter plate **36** also has a lower mounting surface **42** (see FIG. 3) that faces away from the driveshaft housing **34**, and to which an upper mounting flange **43** of the lower gearcase **32** is mounted, as will be further described herein below.
- (10) Referring to FIG. 1, the stern drive **10** has a conventional gimbal ring **41** and gimbal housing **44**. The gimbal ring and gimbal housing **41**, **44** facilitate steering motion of the stern drive **10** about an axially extending steering axis **45**, particularly allowing the stern drive **10** to be steered in port and starboard directions with respect to the marine vessel **31**. The gimbal ring and gimbal housing **41**, **44** also allow for trimming motion of the stern drive **10** relative to the marine vessel **31** about a laterally extending trim axis **46**, particularly permitting the stern drive **10** to be trimmed up and

down with respect to the marine vessel **31**. The type and configuration of the gimbal ring and gimbal housing **41, 44** can vary from what is shown and described. Suitable arrangements that facilitate both steering and trimming motion of the stern drive **10** are disclosed in the above-incorporated patents, and thus are not further herein described. See for example U.S. Pat. No. 9,481,439.

(11) Referring to FIG. **1**, the stern drive **10** has includes a trim actuator **48**, which is configured to trim the stern drive **10** up and down relative to the marine vessel **31**, as shown by double-headed arrow **59**. The type and configuration of the trim actuator **48** can vary from what is shown and for example can include any other suitable conventional hydraulic systems and/or one or more conventional electric motors and/or a combination of these, and/or the like for causing the trimming motion shown at arrow **59**. In the illustrated example, the trim actuator **48** is a hydraulic actuator, including port and starboard cylinders **50** and piston rods **52** extending from the cylinders **50**. The outer ends **54** of the port and starboard piston rods **52** are coupled to the respective port and starboard rearward sides of the driveshaft housing **34** at a rearward pivot joint **56**. The forward ends **58** of the port and starboard cylinders **50** are pivotally coupled to the port and starboard sides of the gimbal ring **41** at a forward pivot joint **60**. Extension of the piston rods **52** from the cylinders **50** upwardly trims the upper drive unit **30** and lower gearcase **32** relative to the marine vessel **31**. Retraction of the piston rods **52** into the cylinder **50** downwardly trims the upper drive unit **30** and lower gearcase **32** relative to the marine vessel **31**. The trim actuator **48** further includes a conventional hydraulic pump and associated control valves (not shown) for supplying hydraulic fluid to the port and starboard cylinders **50**. As is conventional, the piston rods **52** are placed under hydraulic pressure so they are caused to extend outwardly towards an outermost position relative to the cylinder **50**, to thereby fully trim the stern drive **10** up relative to the marine vessel **31**. The piston rods **52** are alternately placed under hydraulic pressure so they are retracted inwardly towards an innermost position relative to the cylinder **50** to thereby fully trim the stern drive **10** down relative to the marine vessel **31**.

(12) Referring to FIG. **4**, the lower gearcase **32** supports a pair of counter-rotating dual propeller shafts **62**. As is conventional, the propeller shafts **62** are caused to rotate about a longitudinal propeller shaft axis **47** (see FIG. **1**) by an internal combustion engine and/or electric motor and/or any other suitable drive mechanism (shown schematically at **64** in FIG. **4**). The drive mechanism **64** is operatively coupled to the propeller shafts **62** by a driveshaft **66**, such that operation of the drive mechanism **64** causes rotation of the propeller shafts **62**. The driveshaft **66** includes a longitudinally-extending first driveshaft portion **65** that extends into the driveshaft housing **34**, an axially-extending second driveshaft portion **68** located in the driveshaft housing **34**, and an axially-extending third driveshaft portion **69** located in the lower gearcase **32**. A conventional gearset **63** operably connects the first driveshaft portion **65** to the second driveshaft portion **68**. The second driveshaft portion **68** extends out of the lower mounting flange **38** of the driveshaft housing **34**, as shown in FIG. **3**. The third driveshaft portion **69** extends out of the upper mounting flange **43** of the lower gearcase **32**, as shown in FIGS. **2** and **3**. The outer ends of the second and third driveshaft portions **68, 69** are splined, and are operably coupled together by an internally splined connector sleeve **73**, such that rotation of the second driveshaft portion **68** causes rotation of the third driveshaft portion **69**. The axially-extending third driveshaft portion **69** is operably coupled to the propeller shafts **62** for example by another conventional bevel gearset **71**, such that rotation of the third driveshaft portion **69** causes counter-rotation of the propeller shafts **62**. Thus, operation of the drive mechanism **64** causes rotation of first, second and third driveshaft portions **65, 68, 69**, which in turn causes counter-rotation of the dual propeller shafts **62**.

(13) Counter-rotating propellers **70** are located on the outer end of the dual propeller shafts **62**, forwardly of the lower gearcase **32**. The propellers **70** are caused to rotate by rotation of the propeller shafts **62**, which thereby creates a propulsive force on the stern drive **10** and propels the marine vessel **31** in the body of water. This is conventionally referred to in the art as a “tractor-

type” stern drive arrangement, wherein the propellers **70**, which are often referred to as “pulling propellers”, are located on the forward side of the lower gearcase **32**, and wherein operation of the stern drive **10** in a forward gear causes the propellers **70** to effectively pull the marine vessel **31** in the surrounding body of water.

(14) Referring to FIG. **3**, the lower mounting surface **42** of the adapter plate **36** has a recess **74** into which the upper mounting flange **43** of the lower gearcase **32** is nested. Thus the outer peripheral sidewall **76** of the upper mounting flange **43** faces the inner peripheral sidewall **77** of the recess **74**. The upper mounting flange **43** is fastened to the lower mounting surface **42** of the adapter plate **36**, at least in part, by (A) a forward mounting joint and (B) a relatively more robust (i.e. mechanically stronger and thus more resistant to failure) rearward mounting joint. In particular, the forward mounting joint is formed by port and starboard fasteners **80** that extend through the port and starboard through-bores **39** in the upper mounting flange **43** and into threaded engagement with port and starboard threaded holes **82** in the adapter plate **36**, respectively. The rearward mounting joint is formed by relatively more robust port and starboard threaded studs **86** that extend from the lower mounting surface **42** of the adapter plate **36** and through port and starboard through-bores **88** in the upper mounting flange **43**. Fastener nuts **90** and locking washers **95** are threaded onto the ends of the port and starboard threaded studs **86** to thereby fasten the lower gearcase **32** to the adapter plate **36**.

(15) Referring to FIGS. **2** and **3**, the adapter plate **36** is fastened to the driveshaft housing **34**, in particular, via threaded studs **92** that extend from a raised pedestal **93** on the lower mounting flange **38** and into corresponding through-bores **94** in the adapter plate **36** and alternately from the adapter plate **36** into corresponding through-bores **94** in the raised pedestal **93**. Fastener nuts **96** are threaded onto the ends of the threaded studs **92** and registered in sunken recesses on through-bores **94** to thereby secure the adapter plate **36** to the driveshaft housing **34**. An additional threaded stud **98** located rearwardly of the threaded studs **92** extends from the upper mounting surface **40** of the adapter plate **36** and is engaged in a threaded bore **100** in the driveshaft housing **34** to further fasten the components together. The manner in which the adapter plate **36** is fastened to the driveshaft housing **34** can vary from what is shown and described.

(16) Referring to FIGS. **2** and **3**, the adapter plate **36** has a centrally-located driveshaft passageway **102** in which the second and third driveshaft portions **68**, **69** are located. The noted forward mounting joint is disposed longitudinally and laterally alongside the driveshaft **66**. In particular, the forward port and starboard threaded holes **82** in the adapter plate **36** and the respective fasteners **80** are diametrically opposed to each other relative to the passageway **102**, i.e., on the port and starboard sides of the driveshaft **66**, respectively. The adapter plate **36** also has a second passageway **104** through which exhaust gases from the stern drive **10** are conveyed from the upper drive unit **30** to the lower gearcase **32**. The noted robust rearward mounting joint is located rearwardly of the driveshaft **66** and rearwardly of the second passageway **104**. In particular, the rearward port and starboard threaded studs **86** and port and starboard through-bores **88** are located rearwardly of the second passageway **104**, and on the port and starboard sides of the second passageway **104**, respectively. A third passageway **106** in the adapter plate **36** is located forwardly of the driveshaft **66** and is for conveying cooling water through the stern drive **10**, in a conventional arrangement, particularly from an inlet **109** (see FIG. **1**) on the lower gearcase **32** to an inlet **111** (see FIG. **3**) formed in the lower mounting flange **38** of the driveshaft housing **34**. The third passageway **106** is located forwardly of the forward mounting joint.

(17) Referring to FIGS. **1-3**, the lower gearcase **32** has a trailing end surface **118** that is angled relative to (i.e., extends transversely to) the lower mounting surface **42** and further has a trailing end surface **112** that extends generally perpendicularly to the lower mounting surface **42**. The lower mounting surface **42** and trailing end surface **112** meet at a rounded corner. There is a three-dimensional space or gap **120** (see FIG. **4**) located (longitudinally, axially, and transversely) between the trailing end of the lower mounting surface **42** and the angled trailing end surface **118**.

The angled trailing end surface **118** and the gap **120** facilitate pivoting movement of the lower gearcase **32** relative to the upper drive unit **30**, as shown in FIG. 5, and subsequent uncoupling and separation of the lower gearcase **32** from the upper drive unit **30**, as shown in FIG. 6, as will be further described herein below.

(18) FIGS. 4-6 depict an initial impact of the stern drive **10** on an underwater obstruction **108**, particularly as the stern drive **10** and marine vessel **31** are moving forwardly. The type and configuration of the underwater obstruction **108** can vary, and for example could include a reef, bedrock, and/or any other sizeable underwater impediment. FIGS. 4-6 depict, impact of the nose cone **110** of the dual propeller shafts **62** on the underwater obstruction **108**; however the concepts of the present disclosure are also applicable in other collisions, for example anywhere along the leading edge of the lower gearcase **32**.

(19) Referring to FIG. 4, the initial impact with the underwater obstruction **108** causes the stern drive **10** to pivot about the trim axis **46**, as shown at arrow **114**, which pivoting movement is permitted by an outward extension movement of the piston rods **52** from the cylinders **50**.

(20) Referring to FIG. 5, either simultaneously upon impact and/or during or after the above-noted pivoting movement of the lower gearcase **32** shown in FIG. 4, the forward mounting joint is caused to fail, for example by shearing of the fasteners **80** and/or stripping of the threaded connection between the fasteners **80** and the corresponding holes **82**. Failure of the forward mounting joint will typically occur when the underwater obstruction **108** is impacted with significant force, for example a force large enough to have otherwise damaged the stern drive **10**, but for the noted failure of the forward mounting joint. As described herein above, the rearward mounting joint is more robust than the forward mounting joint and is thus it is configured to remain intact even when the forward mounting joint is caused to fail, which facilitates subsequent pivoting movement of the lower gearcase **32** about a lateral pivot axis **115**, as shown in FIG. 5. In particular, the lateral pivot axis **115** is defined by the port and starboard threaded studs **86**. The port and starboard threaded studs **86** are more robust than the port and starboard fasteners **80** (e.g. have larger size and/or are made of more resilient material, for example a stronger metal) and less susceptible to failure than the port and starboard fasteners **80**. Also, location of the port and starboard studs **86** rearwardly of the driveshaft **66** and rearwardly of the port and starboard fasteners **80** is further from the point of impact on the forward side **18** of the stern drive **10**, and thus causes the rearward mounting joint to be less likely to fail under the initial impact force of the lower gearcase **32** striking the underwater obstruction **108**. As mentioned, failure of the forward mounting joint upon or after impact, advantageously results in a pivoting movement of the lower gearcase **32** with respect to the upper drive unit **30**, particularly about the rearward mounting joint, and more particularly about the lateral pivot axis **115** defined by the port and starboard threaded studs **86**.

(21) Referring to FIG. 5, shearing of the forward mounting joint and retention of the rearward mounting joint facilitates pivoting of the lower gearcase **32** until the trailing end surface **118** impacts the trailing end of the lower mounting surface **42**, which impact force causes the rearward mounting joint to fail, and thus results in complete separation of the lower gearcase **32** from the upper drive unit **30**, as shown in FIG. 6. That is, the rearward mounting joint is sized and located such that the force of impact between the lower gearcase **32** and the adapter plate **36** along the trailing end surface **118** shears the rearward port and starboard threaded studs **86** and/or strips the threaded connection between the studs **86** and the through-bores **88**, and/or otherwise causes the rearward mounting joint to fail, thus resulting in separation of the lower gearcase **32** from the upper drive unit **30**. That is, the size and location of the studs **86** is such the studs are designed to fail under the force of impact shock that occurs between the lower gearcase **32** and trailing end of the adapter plate **36**. The internally splined connector sleeve **73** facilitates separation of the outer ends of the second and third driveshaft portions **68**, **69** when the noted rearward mounting joint fails.

(22) Advantageously, because the trailing end of the adapter plate **36** is located above and rearwardly of the rearward mounting joint, upon failure of the rearward mounting joint the lower

gearcase **32** pivots about the rounded corner between the lower mounting surface **42** and trailing end surface **112**, and is forced downwardly and rearwardly of the remainder of the stern drive **10**, as shown in FIG. **6**, thus preventing it from traveling upward out of the body of the water. This causes a safer separation of the lower gearcase **32** from the rest of the stern drive **10** and advantageously protects the upper drive unit **30** of the stern drive **10** from being damaged by the impact with the underwater obstruction **108**. It also prevent the lower gearcase **32** from moving upwardly out of the body of water and potentially injuring someone.

(23) The present disclosure thus provides an improved stern drive **10** wherein the upper drive unit **30** and the lower gearcase **32** are specially configured such that when the forward side **18** of the stern drive **10** impacts an underwater obstruction **108**, particularly along the lower gearcase **32**, the lower gearcase **32** is caused to rearwardly pivot relative to the upper drive unit **30** until the trailing end surface **118** of the lower gearcase **32** impacts the lower mounting surface **42**, which thereby causes the lower gearcase **32** to completely uncouple from the upper drive unit **30**. The upper drive unit **30**, the lower gearcase **32**, and the trim actuator **48** are specially configured such that when the forward side **18** of the stern drive **10** impacts the underwater obstruction **108**, particularly along the lower gearcase **32**, the upper drive unit **30** is initially caused to trim relative to the marine vessel **31** and the lower gearcase **32** is caused to pivot relative to the upper drive unit **30**, until the trailing end surface **118** impacts the lower mounting surface **42**, which thereby causes the lower gearcase **32** to completely uncouple from the upper drive unit **30**. When the forward side **18** of the stern drive **10** impacts the underwater obstruction **108**, particularly along the lower gearcase **32**, the upper drive unit **30** is first caused to trim up relative to the marine vessel **31**, which in turn causes the piston rods **52** to extend outwardly to the outermost position, and thereafter the trailing end surface **118** impacts the lower mounting surface **42**, which causes the lower gearcase **32** to completely uncouple from the upper drive unit **30**. Advantageously the lower mounting surface **42** extends rearwardly of the lower gearcase **32**, so that when the trailing end surface **118** impacts the lower mounting surface **42**, the lower gearcase **32** is forced downwardly relative to the marine vessel **31** and more specifically is prevented from moving upwardly out of the water.

(24) In the above description, certain terms have been used for brevity, clarity, and understanding. No unnecessary limitations are to be inferred therefrom beyond the requirement of the prior art because such terms are used for descriptive purposes and are intended to be broadly construed. The different assemblies described herein may be used alone or in combination with other assemblies. It is to be expected that various equivalents, alternatives and modifications are possible within the scope of the appended claims.

Claims

1. A marine drive for propelling a marine vessel in water, the marine drive comprising: a drive unit having a lower surface, and a gearcase rigidly coupled to the drive unit, the gearcase having a trailing end surface that is angled relative to the lower surface such that a gap is defined between the lower surface and the trailing end surface, wherein the drive unit and the gearcase are configured such that when a forward portion of the gearcase impacts an underwater obstruction, the gearcase is caused to partially uncouple from the drive unit and then rearwardly pivot relative to the drive unit until the trailing end surface closes the gap and impacts the lower surface, which thereby causes the gearcase to completely uncouple and separate from the drive unit.
2. The marine drive according to claim 1, wherein the gearcase is mounted to the drive unit by a forward mounting joint and a trailing mounting joint located rearwardly of the forward mounting joint.
3. The marine drive according to claim 2, wherein the drive unit and the gearcase are configured such that when the gearcase impacts the underwater obstruction, the forward mounting joint is configured to fail, thus permitting the gearcase to rearwardly pivot relative to the drive unit about

the trailing mounting joint until the trailing end surface impacts the lower surface, which thereby causes the trailing mounting joint to fail and allows the gearcase to completely uncouple and separate from the drive unit.

4. The marine drive according to claim 1, further comprising a trim actuator configured to trim the drive unit relative to the marine vessel, wherein the drive unit, the gearcase, and the trim actuator are configured such that when the forward portion of the gearcase impacts the underwater obstruction, the drive unit is caused to trim up relative to the marine vessel.

5. The marine drive according to claim 4, wherein the trim actuator is a hydraulic actuator comprising a cylinder and a piston rod extending from the cylinder, wherein the piston rod is extended outwardly relative to the cylinder to thereby trim the marine drive up relative to the marine vessel and wherein the piston rod is retracted inwardly relative to the cylinder to thereby trim the marine drive down relative to the marine vessel.

6. The marine drive according to claim 5, wherein when the forward portion of the gearcase impacts the underwater obstruction, the piston rod extends outwardly.

7. The marine drive according to claim 1, wherein the drive unit extends rearwardly of the gearcase so that when the trailing end surface impacts the drive unit, the gearcase is forced rearwardly and downwardly relative to the marine vessel.

8. The marine drive according to claim 1, wherein the lower surface extends above and rearwardly of the trailing end surface of the gearcase and as such is prevented from traveling upwardly out of the water.

9. The marine drive according to claim 1, wherein the drive unit comprises a driveshaft housing and an adapter plate on the driveshaft housing, wherein the adapter plate is sandwiched between the drive unit and the gearcase.

10. The marine drive according to claim 1, wherein the gearcase is coupled to the drive unit by a forward mounting joint comprising port and starboard fasteners.

11. The marine drive according to claim 10, wherein the gearcase is also coupled to the drive unit by a rearward mounting joint comprising port and starboard threaded studs.

12. The marine drive according to claim 11, further comprising fastener nuts that are threaded onto the port and starboard threaded studs.

13. The marine drive according to claim 11, wherein the port and starboard threaded studs are and on port and starboard sides of the marine drive, respectively.

14. The marine drive according to claim 11, wherein the port and starboard threaded studs define a pivot axis about which the gearcase is caused to rearwardly pivot relative to the drive unit when the forward portion of the gearcase impacts the underwater obstruction.
